

June 26, 2026 Genetics

Genome Informatics

The Common Infrastructure of Modern Genetics

Read → Annotate → Compare → Expand → **Predict with AI**

Hiroshi Mori
Genome Diversity Lab.
NIG

Today's Lecture

1

Why Genome Informatics?

*From one gene to the
whole genome*

2

Assembly & Annotation

*Sequencing →
sequence → gene
models*

3

Comparative Genomics & Genome Variation

*Orthologs, paralogs &
pan-genomes*

4

Beyond Model Organisms

*Metagenomics, aDNA
& the dark genome*

5

AI & the Future

*DNA language models
& what comes next*

Before We Begin

How well do we know the genome of your research organism?

Complete
chromosome-level
assembly?

Draft genome with
many gaps?

No reference genome
exists?

Part 1

Why Genome Informatics?

*"Genetics used to ask: what does this gene do?
Now it asks: what does the entire genome mean?"*

The Revolution in Genetics

30 Years Ago

- Find one gene of interest
- Create knockout / mutant
- Observe phenotype
- Publish one gene paper
- Years per gene



Today

- Read entire genome
- Survey whole transcriptome
- Resolve single cells
- Compare 10s of species
- Predict with AI

The Genomics Revolution: A Timeline

2001



Human Genome Project

*\$3 billion
13 years*

2005



NGS Revolution

*454 & Illumina
short reads*

2011



Long-Read Sequencing

*PacBio
Nanopore*

2015



Single-Cell Genomics

*scRNA-seq
spatial*

2020–



AI & Foundation Models

*Evo
AlphaFold*

Cost of sequencing a human genome: \$3B (2001) → \$100 (2026)

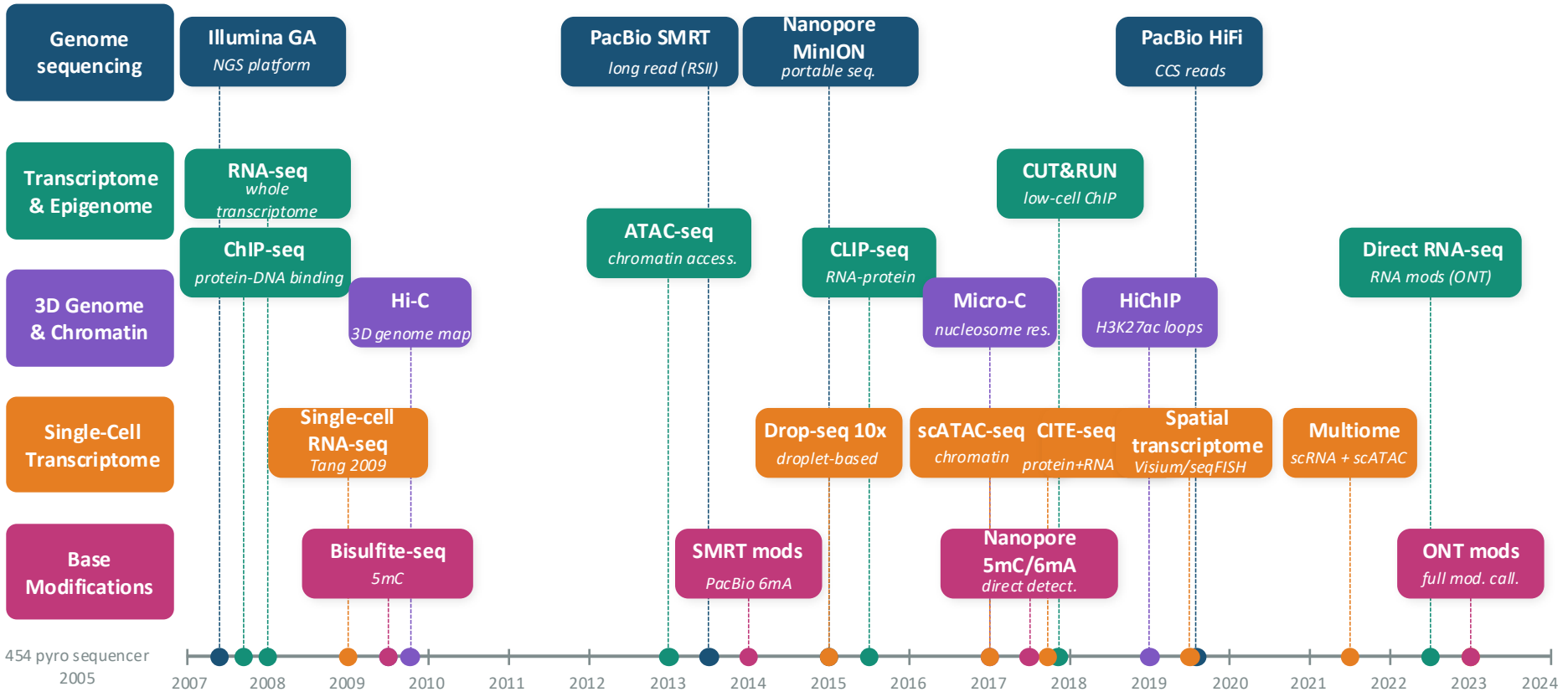
Representative Next Generation Sequencer (NGS) in June 2026

Sequencer Name	Specific property	Read length (base)	Read number / run
ABI 3730xl (not NGS)	Sanger	500–1000	384
illumina NextSeq 2000	Bridge PCR	150	1,800,000,000
illumina NovaSeq X	Bridge PCR	150 or 250	25,000,000,000
Element Biosciences AVITI	Rolling circle amplification	150 or 300	2,000,000,000
PacBio Revio	Single molecule	Average 20,000	3,000,000
Nanopore MinION, GridION, PromethION	Single molecule	Average 10,000-30,000	5,000,000 ~ 200,000,000

More and more new sequencers will be coming soon!

From Reading Genomes to Reading Everything

The Expansion of Next-Generation Sequencing: 2007 → Present



NGS transformed genomics from a single assay (DNA sequencing) into a universal platform for reading genome, epigenome, transcriptome, 3D structure, and chemical modifications — in bulk and at single-cell resolution

What Does Genome Informatics Do?

Genome informatics converts raw sequence data into biological knowledge

Biological

Gene structure
Regulatory elements
Variant effects
Evolutionary history

Computational

Algorithms & data structures
Statistical models
Machine learning
Database design

Applied

Human disease genes
Crop improvement
Drug targets
Conservation biology

Part 2

Genome Assembly & Annotation

"Sequencing gives you letters. Informatics gives them meaning."

From Sanger to Nanopore: Why Read Length Matters

Sanger

1980s–2005

Read length

~1 kb

Accuracy

99.99%

*Gold standard
for single loci*

Illumina

2007–now

Read length

150–300 bp

Accuracy

99.9%

*High throughput
low cost*

PacBio HiFi

2019–now

Read length

10–25 kb

Accuracy

99.9%

*Long + accurate
resolution of repeats*

Oxford Nanopore

2014–now

Read length

10 kb–4 Mb

Accuracy

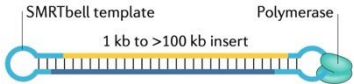
98–99%

*Ultra-long reads
real-time portable*

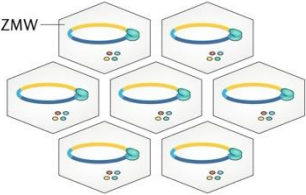
PacBio's sequencing method

a PacBio SMRT sequencing

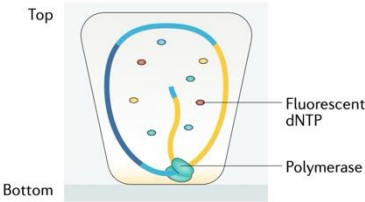
Template topology



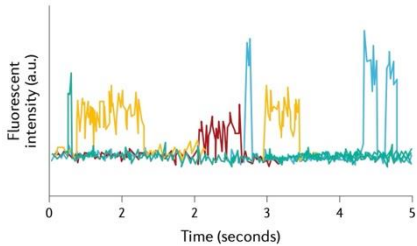
Flow cell (top view)



Single ZMW (cross section)

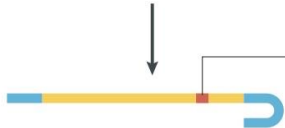
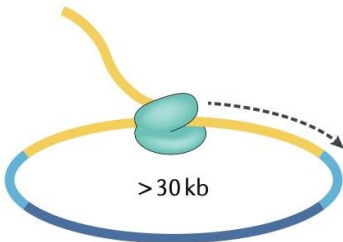


Readout



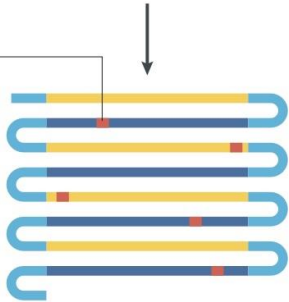
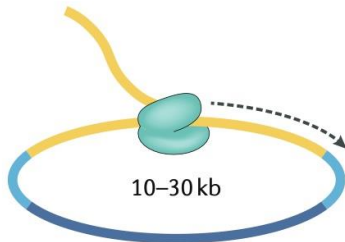
a PacBio CLR and HiFi long reads

CLR sequencing



CLR
Low accuracy

CCS

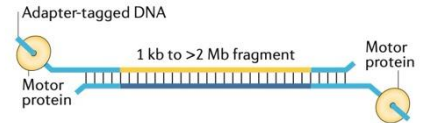


HiFi read
High accuracy

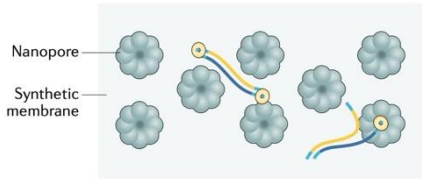
Nanopore's sequencing method

b ONT sequencing

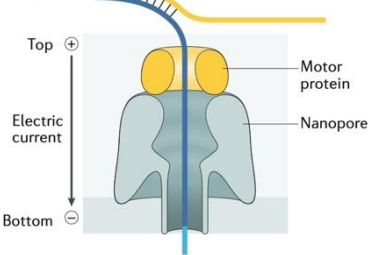
Template topology



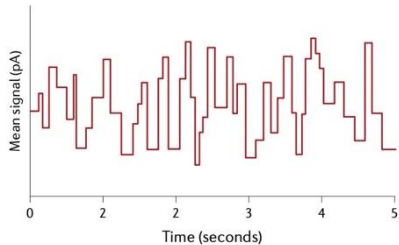
Flow cell (top view)



Single nanopore (cross section)

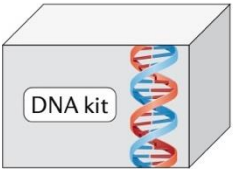


Readout



b ONT long and ultra-long reads

HMW DNA
extraction kit



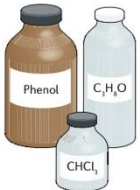
< 100-kb DNA



Long read

Phenol/chloroform,
2-propanol precipitation

or

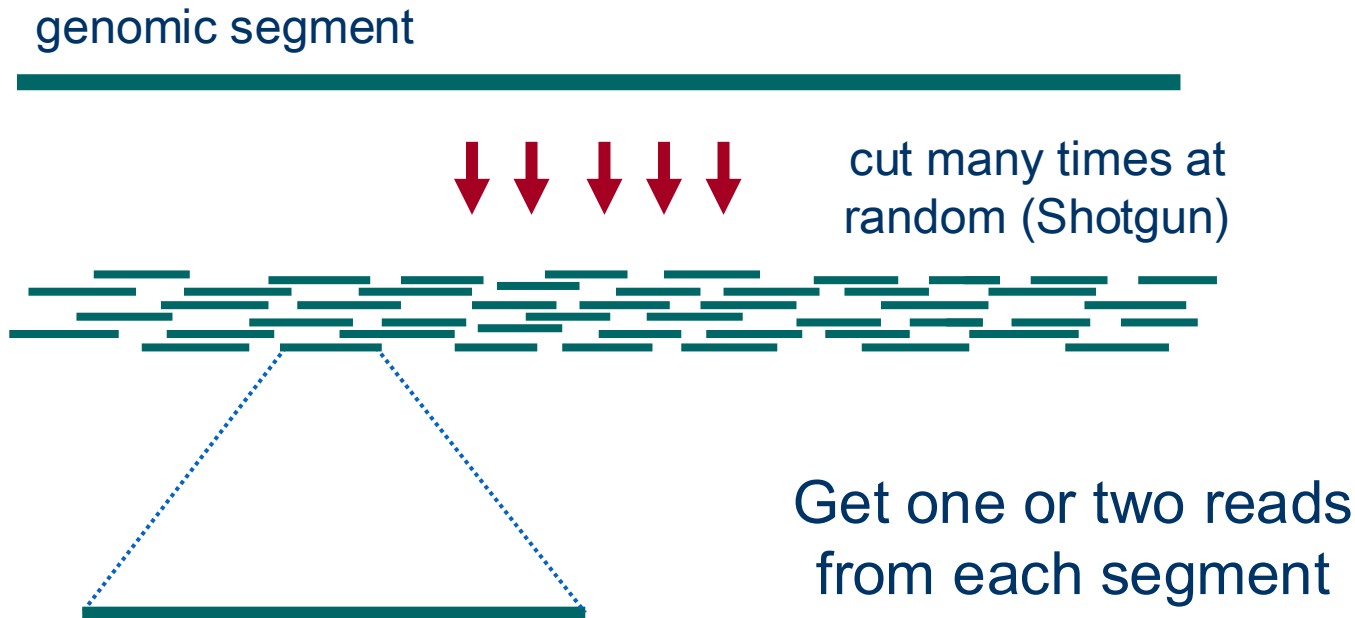


< 5-Mb DNA

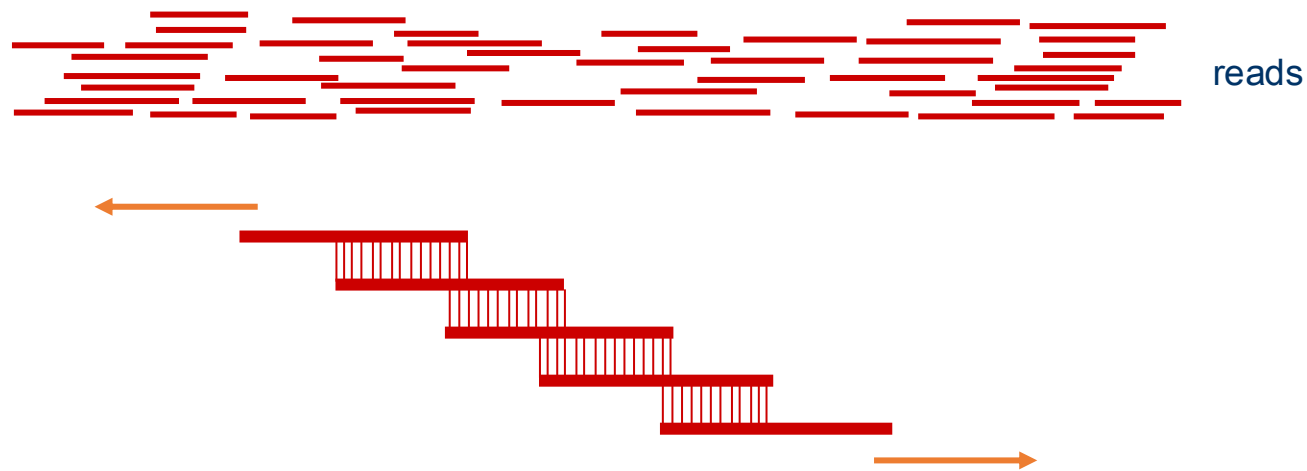


Ultra-long read

Shotgun sequencing



Fragment assembly

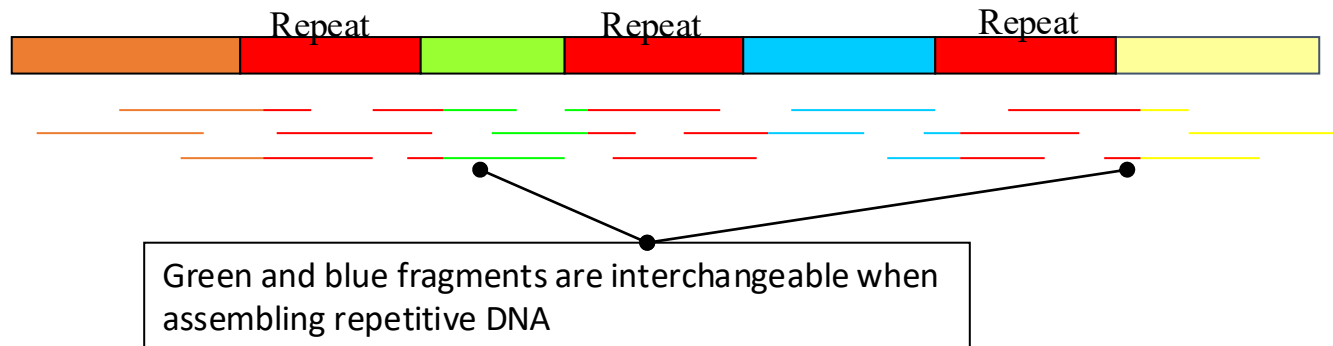


Overlap reads and extend to reconstruct the original genomic region

Overlap-Layout-Consensus approach

Challenges in Fragment Assembly

- Repeats: A **major** problem for fragment assembly
- > 50% of human genome are repeats:
 - over 1 million *A/u* repeats (about 300 bp)
 - about 200,000 LINE repeats (1000 bp and longer)



Long read is preferable!

T2T (Telomere-to-Telomere)

Science 2022

The complete sequence of a human genome

Sergey Nurk^{1,*}, Sergey Koren^{1,*}, Arang Rhie^{1,*}, Mikko Rautiainen^{1,*}, Andrey V. Bzikadze², Alla Mikheenko³, Mitchell R. Vollger⁴, Nicolas Altemose⁵, Lev Uralsky^{6,7}, Ariel Gershman⁸, Sergey Aganezov⁹, Savannah J. Hoyt¹⁰, Mark Diekhans¹¹, Glennis A. Logsdon⁴, Michael Alonge⁹, Stylianos E. Antonarakis¹², Matthew Borchers¹³, Gerard G. Bouffard¹⁴, Shelise Y. Brooks¹⁴, Gina V. Caldas¹⁵, Haoyu Cheng^{16,17}, Chen-Shan Chin¹⁸, William Chow¹⁹, Leonardo G. de Lima¹³, Philip C. Dishuck⁴, Richard Durbin²¹, Tatiana Dvorkina³, Ian T. Fiddes²², Giulio Formenti^{23,24}, Robert S. Fulton²⁵, Arkarachai Fungtammasan¹⁸, Erik Garrison^{11,26}, Patrick G.S. Grady¹⁰, Tina A. Graves-Lindsay²⁷, Ira M. Hall²⁸, Nancy F. Hansen²⁹, Gabrielle A. Hartley¹⁰, Marina Haukness¹¹, Kerstin Howe¹⁹, Michael W. Hunkapiller³⁰, Chirag Jain^{1,31}, Miten Jain¹¹, Erich D. Jarvis^{23,24}, Peter Kerpedjiev³², Melanie Kirsche⁹, Mikhail Kolmogorov³³, Jonas Korlach³⁰, Milinn Kremitzki²⁷, Heng Li^{16,17}, Valerie V. Maduro³⁴, Tobias Marschall³⁵, Ann M. McCartney¹, Jennifer McDaniel³⁶, Danny E. Miller^{4,37}, James C. Mullikin^{14,29}, Eugene W. Myers³⁶, Nathan D. Olson³⁶, Benedict Paten¹¹, Paul Peluso³⁰, Pavel A. Pevzner³³, David Porubsky⁴, Tamara Potapova¹³, Evgeny I. Rogae^{6,7,39,40}, Jeffrey A. Rosenfeld⁴¹, Steven L. Salzberg^{9,42}, Valerie A. Schneider⁴³, Fritz J. Sedlazeck⁴⁴, Kishwar Shafin¹¹, Colin J. Shew²⁰, Alaina Shumate⁴², Yumi Sims¹⁹, Arian F. A. Smit⁴⁵, Daniela C. Soto²⁰, Ivan Sović^{30,46}, Jessica M. Storer⁴⁵, Aaron Streets^{5,47}, Beth A. Sullivan⁴⁸, Françoise Thibaud-Nissen⁴³, James Torrance¹⁹, Justin Wagner³⁶, Brian P. Walenz¹, Aaron Wenger³⁰, Jonathan M. D. Wood¹⁹, Chunlin Xiao⁴³, Stephanie M. Yan⁴⁹, Alice C. Young¹⁴, Samantha Zarate⁹, Urvashi Surti⁵⁰, Rajiv C. McCoy⁴⁹, Megan Y. Dennis²⁰, Ivan A. Alexandrov^{3,7,51}, Jennifer L. Gerton¹³, Rachel J. O'Neill¹⁰, Winston Timp^{8,42}, Justin M. Zook³⁶, Michael C. Schatz^{9,49}, Evan E. Eichler^{4,24,†}, Karen H. Miga^{11,†}, Adam M. Phillippy^{1,†}

¹⁻⁵¹ Affiliations are listed at the end

^{*} Equal contribution

[†] Corresponding authors: Evan E. Eichler (eee@gs.washington.edu); Karen H. Miga (khmiga@ucsc.edu); Adam M. Phillippy (adam.phillippy@nih.gov)

Abstract

In 2001, Celera Genomics and the International Human Genome Sequencing Consortium published their initial drafts of the human genome, which revolutionized the field of genomics. While these drafts and the updates that followed effectively covered the euchromatic fraction of the genome, the heterochromatin and many other complex regions were left unfinished or erroneous. Addressing this remaining 8% of the genome, the Telomere-to-Telomere (T2T) Consortium has finished the first truly complete 3.055 billion base pair (bp) sequence of a

Genome Annotation: Sequence → Knowledge

A genome sequence without annotation is like a book in an unknown language

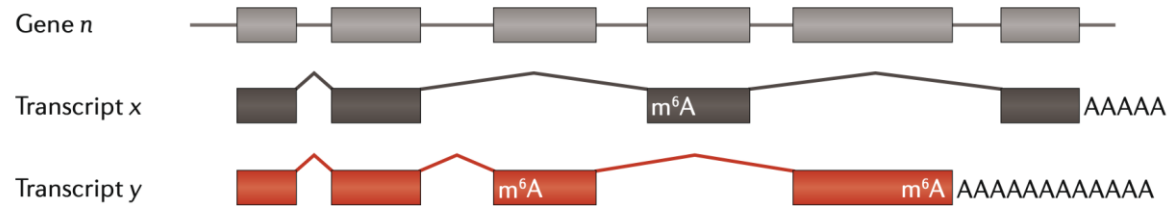
Structural Annotation

- Ab initio gene prediction
- **RNA-seq** data integration
- Splice site identification
- UTR and isoform resolution

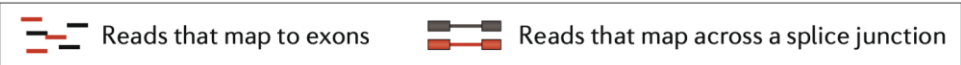
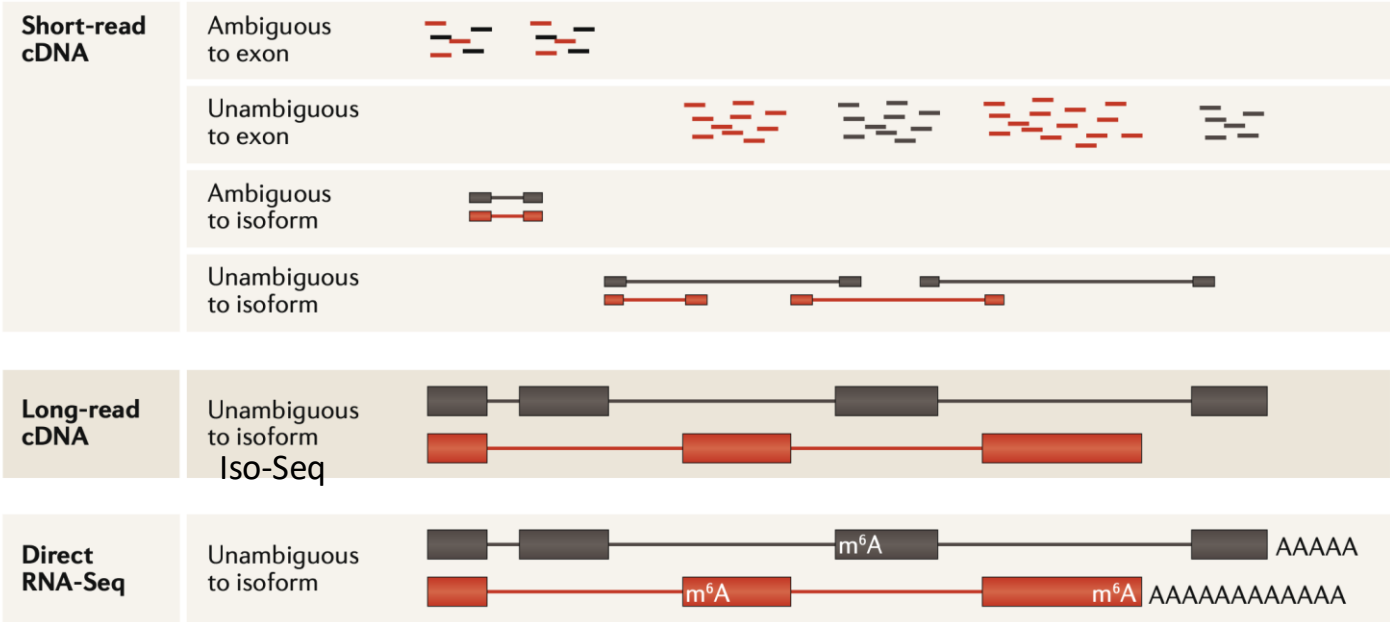
Functional Annotation

- Protein domain search
- GO term assignment
- KEGG pathway mapping
- Homology-based inference

Bulk RNA-seq ^c



Transcriptome mapping



Structural Annotation

Gene Prediction in Eukaryotes

Core Question

Where are the genes, and what is their exon–intron structure?

Why Eukaryotes Are Hard

- Introns interrupt every gene
- Genes = only ~2% of genome
- 1 gene → many splice isoforms
- Repeats confound prediction

AUGUSTUS

Ab initio

BRAKER3

Evidence-guided

AUGUSTUS

Ab initio gene predictor (Stanke et al., 2004–present)

How it works

Generalized HMM (gHMM) that models exons, introns, and intergenic regions as distinct states.

Scores every possible gene parse
→ outputs the highest-probability structure.

Species training

Must be trained per species: exon lengths, codon bias, splice-site motifs.

~260 pre-trained species included. New species → use BUSCO gene set.

Strengths / Limits

- ✓ Works without RNA-seq
- ✓ Accepts RNA-seq hints
- ✓ Predicts alternative isoforms

✗ Needs good training data; UTR prediction unreliable

BRAKER3/4

Evidence-guided: RNA-seq + protein homology (Gabriel et al., 2023)

Pipeline (6 steps)

- ① STAR → align RNA-seq
- ② StringTie2 → assemble
- ③ miniprot → map proteins
- ④ GeneMark-ETP → self-train
- ⑤ AUGUSTUS → predict
- ⑥ TSEBRA → filter & merge

Inputs / Outputs

Inputs

- Genome (FASTA)
- RNA-seq reads
- Protein DB (OrthoDB)

Output

- GFF3 annotation
- CDS + protein FASTA

Strengths / Limits

- ✓ Fully automated
- ✓ Self-trains on target genome
- ✓ RNA-seq + proteins combined
- ✓ Best practice for new genomes

✗ RNA-seq required; lowly expressed genes missed

Workflow: Genome assembly → BRAKER3 → EvidenceModeler → BUSCO completeness check → Manual curation

Functional Annotation

Assigning biological meaning to predicted gene models

~40–60%

of genes in a new
eukaryote: unknown function

1 Sequence Similarity Search

BLASTp · DIAMOND

Principle

Align predicted proteins vs. curated databases.
Transfer the annotation of the closest homolog
("best-hit transfer").

Key databases

SwissProt — manually reviewed; highest reliability
UniRef90 — clustered; broader coverage
NR — maximum recall, many errors

Thresholds example

E-value $\leq 1 \times 10^{-5}$
Identity $\geq 40\%$
Coverage $\geq 60\%$

⚠ Mis-annotations in source DBs propagate automatically. Always prefer SwissProt over NR.

✓ Run DIAMOND vs SwissProt first — fast, reliable, and enough for most genes.

2 Domain & Family Analysis

InterProScan · HMMER

Principle

Protein domains are the functional units of evolution. Profile HMMs detect function even when BLAST fails ($< 40\%$ identity).

InterProScan

One command scans 14 member databases:
Pfam · PANTHER · PRINTS · SUPERFAMILY
HAMAP · CDD · Gene3D · TIGRFAM ...
→ IPR accessions map directly to GO terms

Output

Domain coordinates in GFF3
GO terms (from IPR mapping)
Family classification

⚠ Domain presence $\neq$ activity.

✓ Run InterProScan first — single most valuable annotation step.

3 Ontology Assignment

eggNOG-mapper · KEGG Kofamskan

Gene Ontology (GO)

Three orthogonal vocabularies:

- Molecular Function (e.g. ATP binding)
- Biological Process (e.g. DNA repair)
- Cellular Component (e.g. nucleus)

eggNOG-mapper

Diamond search vs. eggNOG orthogroups
→ transfers GO + KEGG KO + COG
Output: TSV (GO, KEGG KO, COG per gene)

Kofamskan

HMMER-based HMM search vs. KEGG KO HMM
Output: TSV (KO per gene)

⚠ GO transferred from distant homologs: treat as hypotheses.

Part 3

Comparative Genomics and Genome Variation

"Evolution wrote the annotation — comparative genomics reads it."

Orthologs and Paralogs: The Basics

Ortholog

Genes in different species that diverged by speciation — typically retain same function

Paralog

Genes within a genome that diverged by duplication — may gain new functions

Xenolog

Genes acquired by horizontal gene transfer —
common in prokaryotes

Why This Matters

- Orthologs enable functional transfer:
 - if mouse Brca1 is ortholog of human BRCA1,
 - mouse phenotype informs human disease
- Paralog expansion reveals evolutionary novelty: olfactory receptor families, immune gene diversification
- Correct ortholog/paralog assignment is critical — errors propagate across databases
- Tools: OrthoFinder, OrthoMCL, Ensembl Compara

Genome Variation: Types and Detection

Within-species genetic diversity — the raw material for evolution, disease, and adaptation

~4–5M

variants per human genome vs. reference

SNP / SNV

Single Nucleotide Polymorphism

Most common: ~3–4M per genome

Single base substitution

A → G, C → T, etc.

- rs334 (HbS): Glu6Val in β -globin → sickle cell
- APOE ϵ 4: Arg → Cys increases Alzheimer risk 4x

Detection

Illumina short-read WGS
Genotyping arrays (GWAS)

Databases

dbSNP · gnomAD · 1000 Genomes

Indel

Insertion / Deletion (< 50 bp)

~500K per genome

Short insertion or deletion

Often causes frameshift

- BRCA1 c.5266dupC: frameshift → breast cancer risk
- CCR5- Δ 32: 32 bp deletion → HIV resistance

Detection

Illumina WGS + GATK
Long reads for complex indels

Databases

ClinVar · gnomAD

SV

Structural Variant (≥ 50 bp)

~20K per genome; affect > 10 Mb total

Large deletion, duplication, inversion, translocation

- inv(16): AML-associated chromosomal inversion
- FGFR3 tandem dup → chondrodysplasia

Detection

Long reads (PacBio HiFi, Nanopore)
Linked reads · Hi-C

Databases

dbVar · gnomAD-SV · DGV

CNV

Copy Number Variant

~1K per genome; up to Mb scale

Segment present in $\neq 2$ copies (gain or loss)

- SMN1 deletion → spinal muscular atrophy
- SAMD9L gain → MIRAGE syndrome

Detection

SNP arrays · WGS depth
Long reads (most accurate)

Databases

ClinGen · DECIPHER · gnomAD-SV

Short reads (Illumina) excel at SNPs/indels • Long reads (PacBio/Nanopore) are essential for SVs, CNVs, and repeat-mediated variants

Population Genomics

What can we learn from variation across individuals?

Core Data Structure
VCF (Variant Call Format)

One row per variant,
one column per sample.
Standard for all downstream analysis.

Key Tools

GATK HaplotypeCaller SNP/indel calling

PLINK2 GWAS, LD, PCA

ADMIXTURE Ancestry proportions

ANGSD Low-coverage WGS

SweepFinder2 Selection scans

PSMC / SMC++ Demographic history

Three Major Questions in Population Genomics

GWAS — Genome-Wide Association Study

What is associated with disease or phenotype?

- Genotype 100,000s of individuals at millions of SNPs
- Test each SNP for association with phenotype (logistic/linear regression)
- Manhattan plot: $-\log_{10}(p)$ vs. genomic position

⚠ GWAS finds correlation, not causation. Most hits are in non-coding regions (regulatory variants). Fine-mapping and functional follow-up required.

Example: 200+ loci for type 2 diabetes; FTO locus for BMI (affects enhancer, not FTO gene itself)

Selection Scans

Where has natural selection acted?

- Extended Haplotype Homozygosity (EHH) / iHS: detect recent selective sweeps
- FST: differentiation between populations (local adaptation)
- dN/dS (Ka/Ks): positive vs. purifying selection on coding genes

⚠ Demographic events (bottlenecks, admixture) mimic selection signals. Always compare populations and use multiple statistics.

Example: LCT region in Europeans (lactase persistence) — strongest sweep signal in human genome

Demographic Inference

What is the history of a population?

- Site Frequency Spectrum (SFS): rare vs. common variant ratio reflects bottlenecks
- Linkage Disequilibrium (LD): LD blocks decay at rate $\sim 1/N_e$ per generation
- PSMC / SMC++: infer effective population size (N_e) through time from single genome
- Admixture analysis: proportion of ancestry from different source populations

⚠ All methods assume models. Model misspecification is the main source of error in demographic inference.

Example: PSMC of archaic genomes (Neanderthal, Denisovan) revealed bottleneck and isolation history

The Pan-genome: From One Reference to All Individuals

A single reference genome cannot represent the full genetic diversity of a species

The Problem with a Single Linear Reference

✗ Reference bias

Reads from individuals whose sequence differs from the reference map poorly → variants are missed

✗ Missing sequence

~8% of human sequence was absent from GRCh38; centromeres, segmental duplications, SVs under-represented

✗ Population blind

A single genome (even T2T) cannot capture structural diversity: presence/absence variants, alternative alleles

Pan-genome: Core Concepts

Core genome

Sequences shared by all (or nearly all) individuals of the species

Accessory / dispensable genome

Sequences present in some but not all individuals (PAVs: presence-absence variants)

Species	Individuals	Core	Accessory
Human	47 haplotypes	~3.0 Gb	~200 Mb
Mouse (M. musculus)	9 strains	~2.3 Gb	~300 Mb
Rice (O. sativa)	144 accessions	~320 Mb	~180 Mb
E. coli	>1,000 strains	~2.0 Mb	~18 Mb (open!)

Graph Genome: Encoding All Variation

Linear reference

Single path through genome. Variants described as differences from this path. Reference bias inevitable.

Graph genome

Directed acyclic graph (DAG): nodes = sequence segments, edges = connections. Each individual = a path through the graph.

Tools & Applications

Minigraph-Cactus Build pangenome graph from multiple assemblies (used for human pangenome)

PGGB Variation graph builder; all-to-all alignment approach

vg toolkit Read mapping, variant calling, and genotyping on graph genomes

Giraffe (vg) Fast short-read mapping to pangenome graph

Part 4

Beyond Model Organisms

"We know the genomes of ~0.1% of described species. What about the rest?"

The 99% Problem (Eukaryotes)

~8.7M

Estimated eukaryotic
species on Earth

~2.5M

Described and
named species
In Catalogue of Life
<https://www.catalogueoflife.org/data/metrics>

~23,000

Species with
sequenced genomes
<https://www.ncbi.nlm.nih.gov/datasets/genome/>

~30

Established model
organisms in genetics

How Genome Informatics Reaches the 99%

Environmental DNA (eDNA)

*Species detection from water, soil,
air without capturing organisms*

Metagenomics

*Sequencing all DNA in a sample.
Identify community composition
and gene content*

Metagenome-Assembled Genomes (MAGs)

*Bin and reconstruct genomes of
uncultured organisms from
metagenomes*

Ancient DNA (aDNA)

*Recover genetic information from
archaeological remains — extinct
species, ancient humans*

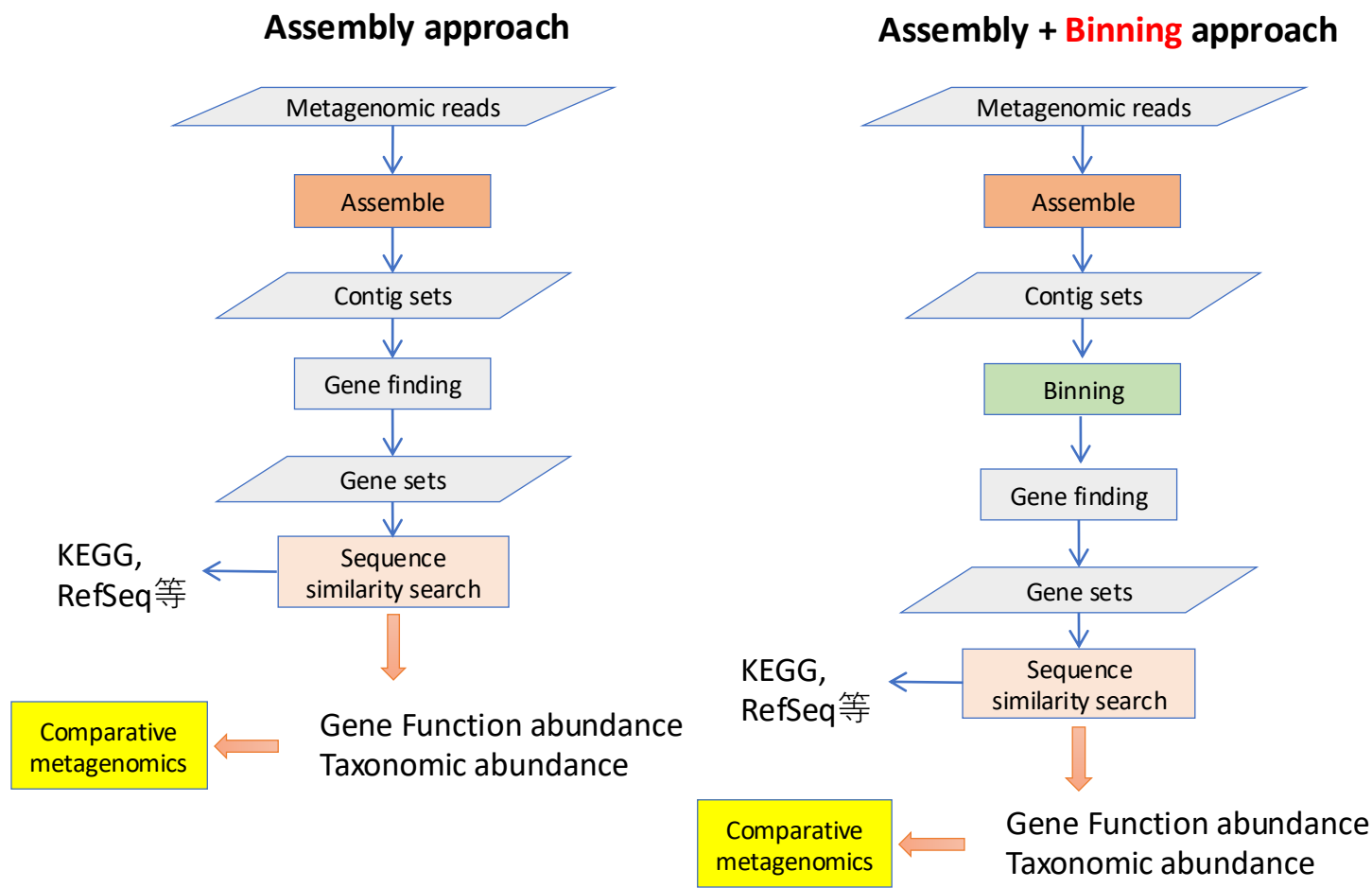
Prokaryotes case

In June 2026

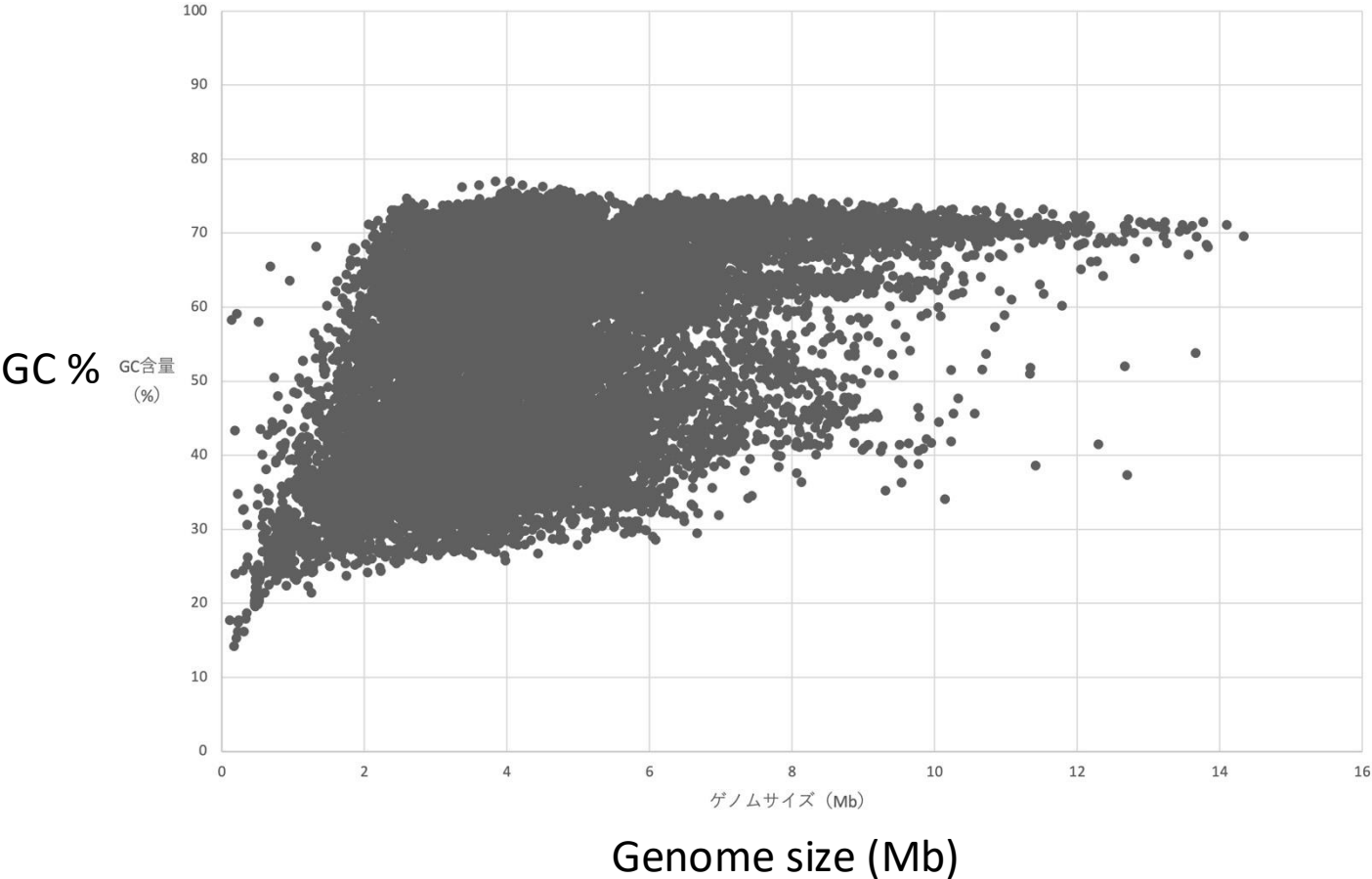
Taxonomic rank	Cultured and validly described taxa	Genome-based putative taxa including uncultured taxa
Phylum	47	186
Class	126	641
Order	256	2,343
Family	669	6,725
Genus	3,989	37,503
Species	22,519	199,923

List of prokaryotic names with standing in nomenclature (<https://lpsn.dsmz.de/text/numbers>)
Genome Taxonomy Database R232 (<https://gtdb.ecogenomic.org/stats>)

Metagenome Assembled Genome (MAG) construction



Metagenome Assembled Genome (MAG) construction



Prokaryotes case

In June 2026

Taxonomic rank	Cultured and validly described taxa	Genome-based putative taxa including uncultured taxa
Phylum	47	186
Class	126	641
Order	256	2,343
Family	669	6,725
Genus	3,989	37,503
Species	22,519	199,923

List of prokaryotic names with standing in nomenclature (<https://lpsn.dsmz.de/text/numbers>)
Genome Taxonomy Database R232 (<https://gtdb.ecogenomic.org/stats>)

Part 5

AI & the Future of Genetics

"Can AI read a genome the way we read a sentence?"

AI & the Future of Genetics

Foundation models are transforming how we read, interpret, and design genomes

1

DNA Language Models

Can we learn the 'grammar' of DNA?

Representative models

DNABERT-2

Nucleotide Transformer

HyenaDNA

Evo / Evo 2

Learn statistical patterns in DNA sequence
→ predict regulatory elements, splice sites,
and variant effects

2

Protein Language Models

Can we predict protein structure from sequence?

Representative models

ESM-2 / ESM-3

AlphaFold 2 / 3

AlphaMissense

ESMFold

Learn evolutionary constraints in protein
sequences → structure prediction, function
annotation, and protein design

3

Sequence-to-Function Models

Can we predict gene expression from DNA sequence?

Representative models

Enformer

Borzoi

AlphaGenome

Sei

CNN/Transformer trained on ENCODE data
→ predict chromatin accessibility, histone
marks, and gene expression from sequence

DNA Language Models — 1/3: The Core Analogy

Why DNA can be treated like natural language

The Language Analogy

Level	Natural Language	DNA
Alphabet	Letters: A–Z	Nucleotides: A / T / C / G
Words	Sequences of letters	k-mers, codons, motifs (e.g. TATAAA)
Grammar	Syntactic rules	Regulatory grammar (splice signals, TF binding motifs)
Context	Surrounding words change meaning	Flanking sequence changes regulatory activity
Long-range dep.	Subject–verb agreement	Enhancer–promoter interactions (10–100 kb apart)

Key Insight

Evolution has written the regulatory code of the genome over billions of years. If a sequence motif is functionally important, it is conserved across species. A model trained to predict masked nucleotides from their context must learn this evolutionary grammar — because the grammar is what makes certain sequences predictable.

DNA Language Models — 2/3: How Training Works

From raw sequence to learned representations

Pre-training Objectives

Masked Language Modeling (MLM)

Used by: DNABERT-2, Nucleotide Transformer



- Model predicts masked bases from flanking context
- Forces learning of motif patterns, codon usage, splice signals

Next-Token Prediction (autoregressive)

Used by: Evo, Evo 2, HyenaDNA



- Model predicts the next base from all previous
- Enables generation of novel sequences

⚠ *The model learns correlations, not mechanisms. Causal interpretation requires experimental follow-up.*

What Does the Model Actually Learn?

Regulatory motifs

TF binding sequences, core promoter elements (TATAAA), polyadenylation signals — learned without any manual labelling

Splice site grammar

GT donor / AG acceptor dinucleotides, branch point sequences — the same rules used by the spliceosome, re-discovered by gradient descent

Codon usage bias

Species-specific codon preferences — learned as part of the 'dialect' of each genome

Evolutionary conservation

Conserved positions (under purifying selection) are easier to predict — model embeddings implicitly encode conservation

✓ No labels needed during pre-training. The supervision signal is the DNA sequence itself — billions of 'correct answers' accumulated by evolution.

DNA Language Models — 3/3: Capabilities

From sequence embedding to functional prediction and design

The Fine-Tuning Paradigm



What Fine-Tuned DNA LMs Can Do

Regulatory element prediction

Predict promoters, enhancers, silencers from sequence alone. Context matters — same motif can be activating or repressive.

Splice site identification

Identify donor/acceptor sites with their exact strength. The surrounding ~200 bp context determines whether splicing occurs.

Variant effect prediction (zero-shot)

Score any SNP by comparing embeddings for REF vs. ALT. Disruptions to learned regulatory grammar produce negative scores.

Transcription factor binding

Predict protein-DNA binding. LMs capture subtle context effects that simple PWM models miss entirely.

Sequence generation and design

Autoregressive models (Evo, Evo 2) generate novel DNA with specified properties. First models to design functional CRISPR-Cas systems de novo.

Cross-species annotation transfer

Pre-train on many species → embed a new genome → regulatory elements cluster correctly without species-specific training.

Protein Language Models — 1/3: The Core Analogy

Proteins as evolutionary language — from sequences to structures and functions

Protein as Language

Level	Natural Language	Protein
Alphabet	26 letters	20 amino acids
Words	Morphemes, words	Motifs, short linear motifs
Grammar	Syntax	Folding rules, contact geometry
Meaning	Semantic content	Biochemical function
Dialect	Regional variation	Species / family variation

Why Protein LMs Are Especially Powerful

- 20 amino acid types vs. 4 nucleotides — richer alphabet
- Protein sequence is MORE conserved than DNA (3rd codon position is silent)
- 250 million known protein sequences vs. ~10,000 fully sequenced genomes
- Protein structures obey strict physical laws — grammar is physics, not convention

Evolution as the Author of Protein Language

Multiple Sequence Alignment — the traditional approach

Align hundreds of homologous sequences. Co-evolving positions (residues that change together across species) indicate direct physical contacts in the 3D structure — 'evolutionary coupling'.

Position i: L A V F [K] G T Y
Position j: R Q M H [E] A S W ← *always change together*
→ **K-E salt bridge in 3D structure!**

What Language Models do instead

Train on 250 million sequences alone (no alignment needed). Transformer attention heads naturally discover co-evolutionary patterns — the model implicitly builds its own MSA.

- ✓ Protein LMs learn 3D contact maps as an emergent property of training on sequence alone — without ever seeing a structure.

Protein Language Models — 2/3: Structure Prediction

From sequence to 3D atomic coordinates — how AlphaFold2 works

AlphaFold2: The Architecture (simplified)

1	Input sequence	Single amino acid sequence (primary structure)
2	MSA	Align homologs from UniRef90 + BFD → co-evolutionary signal matrix
3	Evoformer	48 transformer blocks jointly processing MSA and pairwise residue-residue representation. Attention across sequences (rows) and positions (columns) simultaneously.
4	Structure Module	8 blocks iteratively placing backbone atoms (N, C α , C) in 3D space, constrained by pairwise distances from Evoformer.
5	Recycling	Output structure fed back as input 3 times, refining the prediction iteratively.
6	pLDDT score	Per-residue confidence (0–100). Regions with pLDDT > 90 are as accurate as X-ray crystallography.

 pLDDT < 70 = disordered or unreliable region. Always check confidence scores before using predicted structures.

Before and After AlphaFold2

Before (CASP13, 2018)

- Protein folding = 50-year 'Grand Challenge' of biology
- Best methods correct only ~40% of test targets at CASP13
- Required weeks of compute per protein
- Needed experimental structure as starting point (homology modelling)

After AlphaFold2 (CASP14, 2020)

- AlphaFold2 correct on >90% of CASP14 targets — matched experimental accuracy
- 200 million protein structures predicted in the AlphaFold DB
- Seconds to minutes per protein on a GPU
- Works for completely novel sequences with no known homologs

AlphaFold3 (2024): extended to DNA, RNA, ligands — atomic-resolution prediction of any biomolecular complex

Protein Language Models — 3/3: Beyond Structure

Function prediction, variant scoring, and protein design

The Perplexity Principle — Zero-Shot Function Prediction

A protein LM trained on millions of sequences learns that functional proteins have lower perplexity (are more 'expected') than random sequences or non-functional mutants. This means we can score any variant without training a new model:

$$\text{Score}(\text{variant}) = \log P(\text{variant} \mid \text{protein LM}) - \log P(\text{wild-type} \mid \text{protein LM})$$

Variants that disrupt conserved positions → lower probability → negative score → predicted deleterious

Variant Effect Prediction

AlphaMissense: all 71M possible human missense variants scored for pathogenicity (Science 2023)

EVE: unsupervised model predicts pathogenicity from evolutionary statistics alone

Mutations disrupting conserved positions → lower model probability → predicted deleterious

Function Annotation

Zero-shot GO term prediction: protein LM embeddings cluster by function in latent space

'Dark proteome' annotation: ~40% of proteins have no known function — LMs suggest functional families

Domain detection: domain boundaries appear as abrupt shifts in embedding space

Protein Design

Inverse folding: given a 3D structure, design a sequence that will fold into it (ESM-IF1)

ESM3 (98B params): generate proteins with specified function by conditioning on functional descriptions

Traditional: millions of random variants screened; LM design: start in functional space, test ~100 variants

⚠ Zero-shot scores are predictions. Experimental validation required.

✓ One model predicts structure, variant effects, function, and enables design

Sequence-to-Function Models — 1/3: The Problem

Bridging genotype and phenotype from DNA sequence alone

The Central Problem of Regulatory Genomics

We can read a genome sequence, but cannot predict which genes will be expressed in which cell types, at what level, under what conditions. The regulatory code is largely written in non-coding sequence. 98% of disease-associated variants are in non-coding DNA, yet we lack a systematic way to interpret them.

Why is the regulatory code so hard to read?

Combinatorial

TF binding depends on: motif + neighbouring TF motifs + chromatin state + cell type

Context-dependent

Same motif can be activating in one cell type, repressive in another

Long-range

Enhancers can act on target genes 100 kb–1 Mb away via DNA looping

Emergent

Individual motifs have small effects; combinations create non-linear, cell-type-specific outputs

⚠ ~5% of the human genome is under purifying selection. Most of this is non-coding. Without sequence-to-function models, this 5% is uninterpretable.

The Sequence-to-Function Solution

Core Idea

Train a supervised model on matched (DNA sequence → functional readout) data from ENCODE, GTEx, Roadmap Epigenomics. The model learns f such that $f(\text{sequence}) \approx \text{observed signal across hundreds of cell types simultaneously}$.

Inputs and Outputs

INPUT →

- DNA sequence (196 kb – 1 Mb)
- No cell-type label needed
- No epigenomic data needed

OUTPUT →

- ATAC-seq track: open chromatin
- H3K27ac: active enhancers
- RNA-seq: gene expression levels
- Splice junction usage
- Hi-C contact probabilities

Sequence-to-Function Models — 2/3: Architecture

How Enformer and AlphaGenome go from bases to tracks

Enformer (2021) — the paradigm-setter

1 One-hot encoding

Input: 196,608 bp DNA window. Each base = 4-channel binary vector ([1,0,0,0] for A, etc.)

2 Convolutional stem

7 CNN + pooling layers. Each layer doubles receptive field, halves resolution. Extracts local motifs.

3 Transformer blocks

11 transformer blocks. Self-attention integrates information across the full 196 kb window. Captures long-range enhancer–promoter interactions.

4 Multi-head output

5,313 output tracks — one per ENCODE/CAGE/ATAC experiment. Each track predicts signal at 128-bp bins.

⚠ Enhancers are often >50 kb from target promoters. Models with short receptive fields cannot 'see' the enhancer when predicting at the promoter.

✓ Enformer (196 kb context) was the first model to capture most distal enhancers in one pass. Gene expression prediction improved 2× over CNN-only models.

AlphaGenome (2025–26) — what changed?

Context length

196 kb (Enformer)

→ 1,000,000 bp (1 Mb)

Output resolution

128 bp bins

→ 1 bp (single base)

Modalities

CAGE + ATAC +
H3K27ac

→ + splicing junctions
+ Hi-C contact maps
+ PolyA site usage

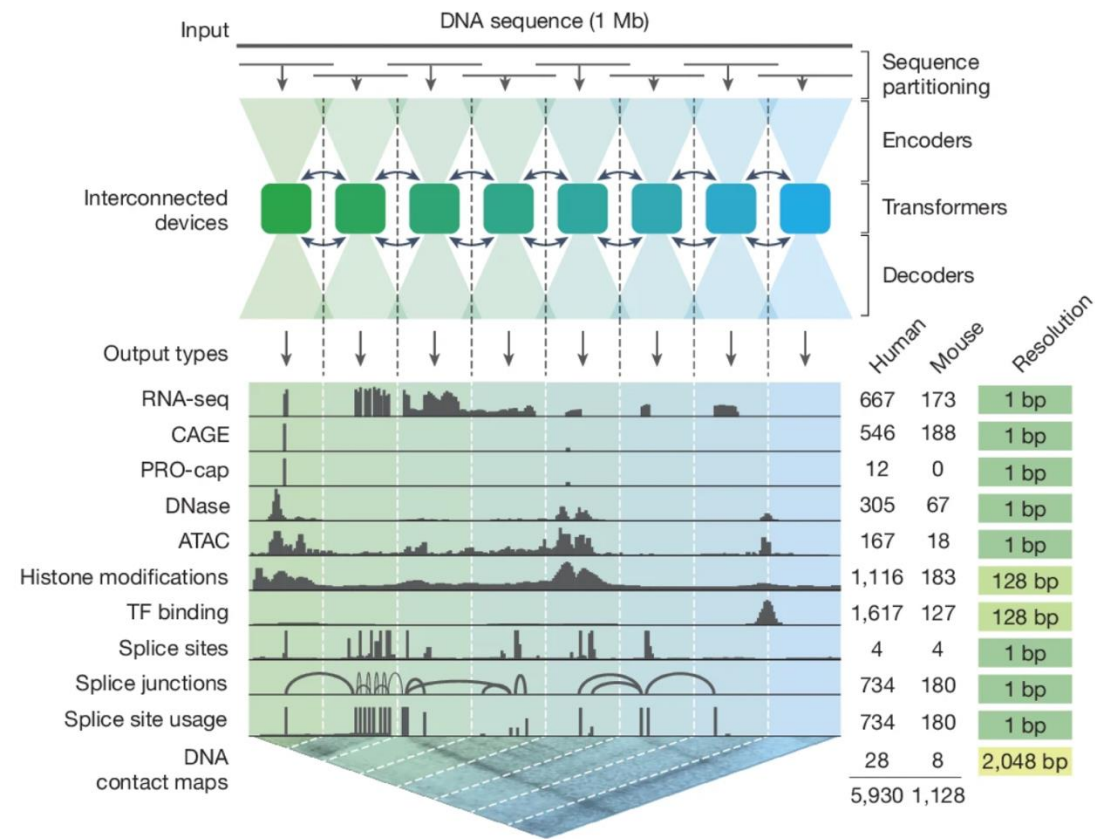
Variant scoring

Compare window
averages

→ Single-base substitution
at exact position

AlphaGenome (Nature 2026): single unified model • all regulatory modalities • single-base resolution • 1 Mb windows

AlphaGenome model structure and training datasets



Modality	Evaluation	Metric	Resolution
Splicing	Splice site classification	auPRC	1 bp
	Splice site usage	Pearson r	1 bp
RNA expression	RNA-seq coverage	Pearson r	1 bp
			32 bp
	RNA-seq gene expr. LFC	Pearson r	Gene
	CAGE coverage	Pearson r	32 bp
DNA accessibility			128 bp
	Alternative poly(A)	Spearman r	Gene
	DNase-seq coverage	Profile JSD	1 bp
		Pearson r	32 bp
Histone modifications			128 bp
	ATAC-seq coverage	Profile JSD	1 bp
		Pearson r	32 bp
			128 bp
TF binding	Histone ChIP-seq coverage	Pearson r	32 bp
			128 bp
DNA contact map	TF ChIP-seq coverage	Pearson r	32 bp
			128 bp
DNA contact map	Orca contact maps	Pearson r	4,000 bp
	Orca contact maps cell type differences	Pearson r	4,000 bp

AlphaGenome input and output

Input

1 million base
DNA sequence:

...
AGCTGCGA
GTGGTATT
ACAGGTCT
AGCGCGAG
TTGAAACA
TAGAGGGG
CGCTCGGT
TATCAACA
...

AlphaGenome

Output

Modalities

1 million base DNA sequence

Gene expression

Accessibility

Histone modifications

TF binding

Splicing

Chromatin contact maps

Thousands of predictions of molecular properties across different cell types

<https://deepmind.google.com/science/alphagenome/>

Sequence-to-Function Models — 3/3: Applications

From regulatory prediction to in silico genetic experiments

What these models enable

A

Non-coding variant effect prediction

Score any SNP or indel by comparing model output for REF vs. ALT sequence. Identifies which non-coding GWAS variants affect gene regulation — the fine-mapping problem.

AlphaGenome identified a cancer-driving insertion near TAL1 that creates a new enhancer — confirmed by ATAC-seq. 0 experiments needed to generate the hypothesis.

B

In silico saturation mutagenesis

Substitute every position in an enhancer with each of the 4 bases. Score each variant with the model. The resulting map shows which positions are essential — a computational regulatory grammar read-out.

Identifies exact TF binding positions within an enhancer without any ChIP-seq. Agrees with experimental footprinting at ~90% specificity.

C

Cell-type-specific regulatory prediction

The model predicts which cell types respond to a given regulatory sequence — without cell-type-specific fine-tuning. Trained simultaneously on 100s of cell types.

Predict which of 200 tissue types will show increased ATAC-seq signal around a SNP. Output: tissue-specific effect sizes, testable as hypotheses.

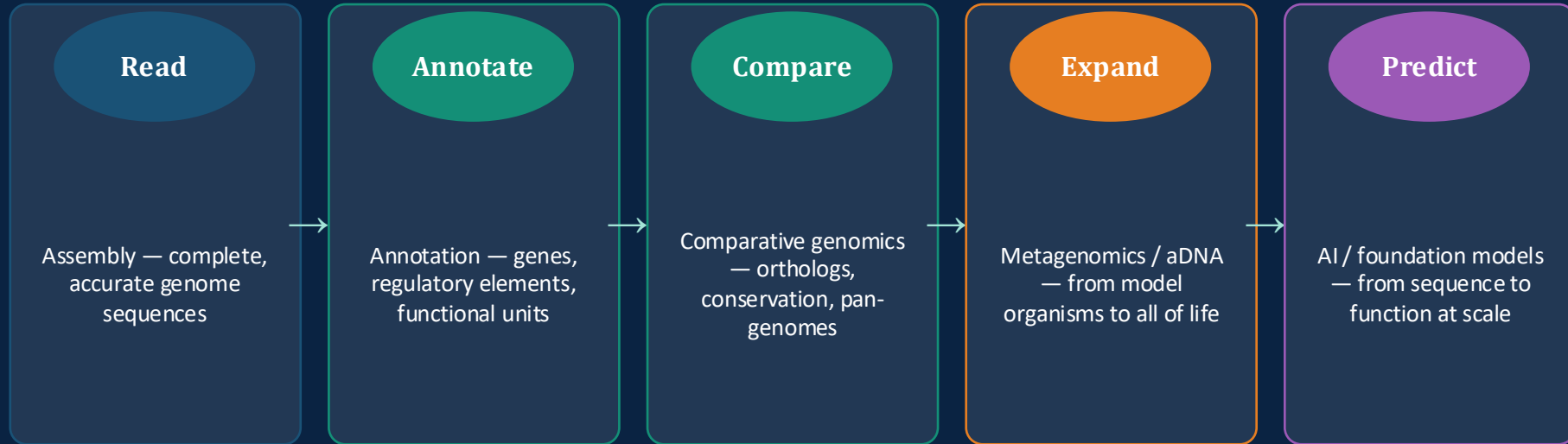
D

Regulatory element design (inverse problem)

Use gradient ascent in sequence space to design enhancers with specified activity profiles: e.g., 'active in liver but not brain'. Model gradients guide the design.

Designed synthetic enhancers with liver-specific activity confirmed experimentally. Random sequence → functional enhancer in 10 rounds of model-guided design.

Genome Informatics: The Unifying Framework



Genome informatics did not just create new tools — it expanded the subject of genetics from a handful of model organisms to the entire tree of life.